

Ádám Porkoláb, PhD

Senior Full Stack Engineer & AI Champion · Java / Spring Boot / Angular

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Senior Engineering Profile
Senior full-stack engineer (Java / Spring Boot / Angular) at Számlázz.hu (Visma group), working on invoicing, tax validation and partner APIs. Internal AI Champion for Agentic Development & Orchestration and for Prompt & Context Engineering. Previously two years at Deutsche Telekom on invoice processing for 13 EU countries. Full details and portfolio at aporkolab.com.

Senior Engineering Scope

Domain	Demonstrated Capability
Full Stack Ownership	Taking features from database to UI independently, navigating unfamiliar codebases
Distributed Systems	Event-driven patterns, Kafka and IBM MQ, resilience (retry/circuit breaker/DLQ)
Legacy Stabilization	Refactoring undocumented systems under production pressure
Code Quality	Clean architecture, testable design, maintainability-first mindset
Remote Collaboration	Async-first workflows, written specs, international team coordination
Observability	Logging strategy, traceability, debugging distributed flows

Engineering Timeline

Senior Full Stack Developer · AI Champion

Budapest, Hungary (Hybrid)

Számlázz.hu (KBOSS.hu Kft., a Visma company)

2026 – present

- Core invoicing services in Java / Spring Boot and Angular, from API design to UI
- Tax and regulatory validation against Hungarian invoicing and reporting rules
- APIs and integrations for partner and accounting systems
- AI Champion, Agentic Development & Orchestration: remit spans specification, development, testing, review and delivery; vendor-neutral coordination of agents, tools and human checkpoints
- AI Champion, Prompt & Context Engineering: responsible for shared, versioned instructions, skills and context patterns; evaluation methods, antipatterns and guidance for teams
- Unified the mail-sending stack
- Code reviews; architecture and API documentation

Open-Source & Independent Projects

Independent

Remote

2020 – present

- Kalliope (2026): reverse-engineered a 2000s Hungarian verse-scansion program from its Delphi binary, rewrote the engine in Java and published it with the original author as co-author (Spring Boot, Angular, Docker; runs in the browser via TeaVM)
- chaos-symphony (2025): event-driven microservices on Java 21, Spring Boot and Kafka, with Saga, Outbox, OpenTelemetry and fault injection

- NyelvSzó: search and editing platform for the Hungarian–English dictionary of linguistics I compiled with Tamás Fekete
- Contributor to miserend.hu (PHP, Angular), the Hungarian Catholic Mass-schedule site

Full Stack Developer (Java / Spring / Angular)

Deutsche Telekom

Pécs, Hungary (Hybrid)

2024 – 2025

- Invoice-processing system for 13 EU countries, each with its own tax rules and validations
- Microservices with Saga and Outbox patterns (Java / Spring Boot, Angular)
- Refactored undocumented legacy monoliths into domain-driven modules
- Async work with German and Hungarian teams on critical production flows

Full Stack Developer (Java / JavaScript / Agile)

Games for Business

Pécs, Hungary

2023

- Developed gamified learning platforms in Java and JavaScript, with PHP (Laminas) integration
- Built responsive UI components that work across devices
- Improved database performance and set up CI/CD workflows
- Pair programming and short iteration cycles in an Agile team

IT Specialist

Baranya County Police Headquarters

Pécs, Hungary

2021 – 2023

- Managed IT infrastructure and network security in a high-security public-sector environment
- Developed internal tools and automation scripts (Java, PHP) alongside a senior colleague
- Helped the in-house developers improve custom tooling and its performance

Software Development Intern (Java/Spring)

InnoTek

Pécs, Hungary

2021

- Java (Spring Boot) and Angular development
- Built MySQL-backed applications under senior developers' mentoring: object-oriented design patterns, testable code

High School Teacher – History & Literature

University of Pécs

Pécs, Hungary

2017 – 2020

- Vocational secondary school of the University of Pécs (social and health care)
- Taught Hungarian language and literature and history; prepared students for the matura and examined them
- Designed curricula, classroom material and e-learning content
- Drafted the school's digital-communication guidelines for parents and students
- Co-authored a study on the 2020 switch to remote teaching in Hungarian schools (Iskolakultúra)
- Before this: Hungarian and history teacher at Testvérvárosok Terei Általános Iskola, Pécs (2016–2017)

University Lecturer (Linguistics)

University of Pécs

Pécs, Hungary

2013 – 2016

- Department of Linguistics, Faculty of Humanities

- Taught Hungarian orthography, text comprehension and communication, and gender-linguistics seminars in BA, MA and teacher-training programmes
- As the doctoral school's student liaison, helped replace its outdated website and represented PhD students on the Doctoral Council

Specialist Librarian (LingBib team)

University of Pécs

Pécs, Hungary

2008 – 2009

- Linguistics Library, Faculty of Humanities
- Answered readers' literature requests across a 15,000-volume specialist collection
- Wrote a plain-language user guide for the library's internal software
- Kept the catalogue database current: new acquisitions and reader records
- Inventoried and catalogued 3,000+ volumes during the summer stocktakes

Tech Stack & Capabilities

Backend

- E-invoicing & Tax-Compliance Systems
- Java 21 & Spring Ecosystem (Boot, MVC, Data, Security)
- Distributed Systems & Event-Driven Architecture
- Apache Kafka, IBM MQ & Messaging Patterns
- Resilience Patterns (Circuit Breaker, Retry, DLQ, Outbox)
- Legacy System Stabilization & Refactoring
- Clean Architecture & Domain-Driven Design
- JPA / Hibernate, PostgreSQL, MySQL
- REST API Design & Documentation (OpenAPI/ Swagger)

Devops

- Docker & Kubernetes
- CI/CD Pipeline Design (Maven, GitHub Actions, Jenkins)
- Google Cloud Platform (GCP)
- Linux System Administration

Writing

- Technical Documentation & Knowledge Architecture
- Architecture Decision Records (ADR) & Tech Blogging

Education

Frontend

- Angular 14+ (TypeScript, RxJS, Signals, Standalone Components)
- Enterprise Component Architecture & State Management
- Responsive UI/UX Implementation
- REST API Integration & Error Handling

Methodology

- Agentic Development & Orchestration (multi-agent workflows, human review)
- Prompt & Context Engineering (instructions, skills, evaluation)
- Async-First Remote Collaboration
- JUnit, Mockito, Testcontainers & Integration Testing
- Observability (OpenTelemetry) & Structured Logging

PhD in Linguistics (Summa Cum Laude, 96%)

2013 – 2018

University of Pécs

- Doctoral School of Linguistics, Applied Linguistics Programme. Supervisor: Dr. habil. József Horváth. Dissertation: an internet-linguistic study of the Hungarian blogosphere by content type and author gender, published as a monograph (2018). Author or co-author of 15 articles, book chapters and reviews in internet linguistics, sociolinguistics and AI in language learning, and of the Hungarian–English and English–Hungarian dictionaries of linguistics (2021).

Frontend Developer, Computer Software Engineering

2021 – 2022

Training360

- Grade A+ (95%). Újratervezés 2.0 IT reskilling programme; admitted among the top 5% of about 10,000 applicants. Nationally recognized “Frontend Developer” and “Full-stack Developer” qualifications.

Java Backend Developer, Computer Software Engineering

2020 – 2021

StudiCore Kft.

- Mentored Java backend course: Java SE, Maven, Git, JPA/Hibernate, Spring, Thymeleaf, MySQL; clean code and SOLID principles.

State-accredited vocational qualification (OKJ), Software Development

2019 – 2020

OKTÁV Továbbképző Központ Zrt.

- National vocational qualification as a software developer, completed alongside full-time teaching.

MA in Hungarian Language and Literature, History (teacher training)

2010 – 2013

University of Pécs

- Graduated with the highest classification (kitűnő). Two-subject teacher-training MA. Thesis: the “language and society” topic in 12th-grade secondary-school textbooks.

BA in Hungarian Language and Literature (History specialization)

2007 – 2010

University of Pécs

- Graduated with the grade jeles (very good).

Languages

- Hungarian (Native)
- English (C1 – Certified)
- German (B2 – Limited working proficiency)

Interests

- **Speculative Fiction Author:** Speculative fiction in English and Hungarian, published internationally; Writers of the Future finalist (2026).
- **System Design & Architecture:** Applying linguistic structuralism to software patterns, API design, and domain modeling.

Publications

- **The Use of Artificial Intelligence in Language Learning (in Hungarian)** (2023)
- **Hungarian–English and English–Hungarian Dictionary of Linguistics (Magyar–angol és angol–magyar nyelvészeti szakszótár)** (2021)
- **Quarantine Pedagogy in Hungarian Public Education: Early Experiences of the Switch to Digital Teaching (in Hungarian)** (2020)

Awards & Certifications

- **2026:** Finalist, Writers of the Future Contest
- **2026:** Highly Commended, erbacce-prize for Poetry
- **2024:** Reactive Programming with Spring Boot (Training360)
- **2016:** ECL Language Exam: English C1 (written and oral)
- **2015:** ECL Language Exam: German B2 (written and oral)

- **From Arkngthand to Wretched Squalor: Fictional place-names in The Elder Scrolls universe (2019)**
- **Web Diary and Blog Business: An Internet-Linguistic Study of the Hungarian Blogosphere by Content Type and Gender (in Hungarian) (2018)**